

Soham Chatterjee

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EDUCATION

Tata Institute of Fundamental Research Integrated PhD in Computer Science	2024–Present
Chennai Mathematical Institute BSc. in Math and Computer Science	2021–2024

AWARDS & FELLOWSHIPS

TCS Research Scholar Program Fellowship Cycle 20, Tata Consultancy Services	2026–2030 Four-year doctoral fellowship
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PUBLICATIONS

- [1] **Soham Chatterjee**, PRAHLADH HARSHA, and MRINAL KUMAR. *Deterministic list decoding of Reed-Solomon codes*. In *Proc. 58th ACM Symp. on Theory of Computing (STOC)*, pages 2040–2051. 2026. [arXiv:2511.05176](#), [eccc:2025/170](#), [doi:10.1145/3798129.3800908](#).

DISSERTATION

MSC Project: Exponential Sums and Weil Bounds: A Survey of Elementary Methods and Decoding Applications
Guide: [Mrinal Kumar](#), STCS [\[PDF\]](#)
May, 2026 [\[Repo\]](#)

RESEARCH INTERESTS

I am broadly interested in Error Correcting Codes, Pseudorandomness, Algebra and Computation, Computational Complexity Theory. My current work is focused on efficient unique and list decoding algorithms of good codes and their local decodability/correctability.

TALKS AND PRESENTATIONS

- **Deterministic List Decoding of Reed-Solomon Codes** with Prahladh Harsha, Mrinal Kumar
 - Talk at CSE Department, IIT Hyderabad
 - Talk at RAAM 2026, IIT Hyderabad
 - Talk at STOC 2026, Salt Lake City, USA
 - Poster at WACT 2026, University of Copenhagen
 - Invited Talk at ACMU Seminar, ISI Kolkata
- **Exponential Sums and Weil Bounds: A Survey of Elementary Methods and Decoding Applications**, TIFR MSc. Project Seminar, 2026 MSc project under Mrinal Kumar

WORKSHOPS AND CONFERENCES

- Pseudorandomness & High Dimensional Expansion, Simons Institute for the Theory of Computing, UC Berkeley, 2026
- Recent Advances in Applied Mathematics (RAAM) 2026, IIT Hyderabad
- Symposium on Theory of Computing (STOC) 2026, Salt Lake City, Utah, USA
- Workshop on Algebraic Complexity Theory (WACT) 2026, University of Copenhagen
- Complexity Update Meeting, 2026, Institute of Mathematical Sciences, Chennai

- Workshop on High Dimensional Expanders, 2025, LMSI
- FSTTCS 2025 Conference, BITS Goa
- HDX and Codes Workshop, 2025, ICTS Bangalore
- FSTTCS 2024 Conference, IIT Gandhinagar
- Quantum Computing Semester, 2024, Chennai Mathematical Institute

PAPER REVIEW

- Reviewer for FSTTCS, 2026.
- Reviewer for ISAAC, 2026.
- Sub-reviewer for STACS, 2026.

RELEVANT COURSEWORK

TIFR	Advance Coding Theory Algebra, Number Theory & Computation Combinatorial Optimization Algorithms Mathematical Foundations in Computer Science	Pseudorandomness Complexity Theory Algorithmic Game Theory Probability Theory
CMI	Computer Science Courses: Expander Graphs & Its Applications Algorithmic Coding Theory Quantum Algorithmic Thinking Parallel Algorithms Theory of Computation Math Courses: Commutative Algebra Rings and Field Theory (Algebra III) Group Theory (Algebra II) Linear Algebra (Algebra I)	Complexity Theory Algebra & Computation Quantum Information Theory Design & Analysis of Algorithms Discrete Mathematics Topology Metric Space (Analysis III) Analysis over Euclidean Space (Analysis II) Real Analysis (Analysis I)

MISCELLANEOUS EXPERIENCE

- Teaching Assistant:
 - Algebra, Number Theory & Computation, 2026 course taken by [Mrinal Kumar](#) in TIFR.
 - Algorithms, 2025 course taken by [T. Kavitha](#) in TIFR.
- Assisted in the selection process for [Vigyan Vidushi](#), TIFR.
- I have volunteered for various outreach activities at TIFR such as [Frontiers of Science](#) and [National Science Day](#) between November 2024 to February 2026 to inspire students to pursue pure science as a vocation and showcases recent scientific advances to the general public.